

many echocardiographers still employ BSA, and presently there is no consensus on a best approach. Further studies comparing alternate methods of aortic indexing are needed to determine the optimal approach.

CONFLICTS OF INTEREST

None.

SUPPLEMENTARY DATA

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.echo.2024.06.009>.

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The Learning Curve for Left Atrial Strain Analysis



Left atrial (LA) strain, in particular, LA reservoir strain, has emerged as an important quantitative tool for the assessment of LA function in patients with heart failure, atrial fibrillation, valvular heart disease, and noninvasive estimation of left ventricular (LV) filling pressures.¹ However, LA deformation analysis by two-dimensional speckle-tracking can be challenging due to (1) thin wall of LA, (2) hypermobile or aneurysmal interatrial septum, and (3) origins of pulmonary veins

and atrial appendage obscuring LA endocardial border definition.² Previous studies have demonstrated learning curves for visual estimation of ejection fraction,³ wall motion analysis during stress echocardiography,⁴ LV global longitudinal strain (GLS),⁵ and right ventricular strain.⁶ The American College of Cardiology, American Heart Association, and American Society of Echocardiography recommend 50 studies to achieve level III training competency in GLS.⁷ Currently, there are no specific recommendations for level of training required for LA strain analysis for independent reporting. This study aims to determine whether a learning curve exists for LA strain analysis to achieve independent competency in accuracy and reproducibility. This is of paramount importance for quality control if LA strain is to be incorporated into routine clinical practice.

This was a retrospective single center study at a tertiary hospital. Exclusion criteria included atrial fibrillation, mitral valve replacement or repair, and suboptimal image quality. Eight independent blinded observers in 3 groups—(1) expert: 2 experts consisting of 1 cardiologist and 1 sonographer with extensive experience in LA strain; (2) competent: 3 experienced observers in LV but not LA strain analysis; (3) novice: 3 observers with no prior experience in strain analysis but who are practitioners of echocardiography—retrospectively analyzed 50 LA strain studies acquired from LA-focused apical 4- and 2-chamber views acquired using Vivid E95 ultrasound system equipped with a M5S 3.5 mHz transducer (GE Vingmed Ultrasound). Biplane LA strain analysis using R-to-R interval was performed offline using dedicated LA strain vendor-specific software (EchoPAC ver. 206, GE Vingmed Ultrasound). This protocol was similar in design to previous GLS and RV learning curve studies.^{5,6} Left atrial strain analysis was performed using a semiautomated computerized algorithm, but manual processing was performed to adjust LA endocardial contours or region of interest width when deemed necessary. Semiautomated tracking was checked by visual inspection in all patients. Manual adjustments were particularly warranted when endocardial borders were interrupted by anatomical structures such as LA appendage and pulmonary veins.

Learning curves were derived by analyzing LA strain in 5 consecutive groups each consisting of 10 patients. Each group's progressive performance in consistency, reproducibility, and time efficiency for LA strain was blinded and independently compared with experts using linear regression analysis with intraclass correlation coefficients (ICCs) and Bland-Altman analysis. An ICC plateau >0.90 was considered equivalent to expert-level competency.

This study demonstrated a minimum of 30 studies were required to achieve expert competency (ICC > 0.90) in LA reservoir strain. **Figure 1** demonstrated both groups achieve similar competency to experts after 30 studies, but beyond this there were still further improvements observed in limits of agreement (LOA) and ICC did not finally plateau until 50 studies. At the commencement of the study, LV strain competent observers demonstrated superior correlation when compared to experts (competent ICC, 0.847; 95% CI, 0.503-0.960), whereas there was a weaker correlation for novice observers (novice ICC, 0.755; 95% CI, 0.281-0.933; **Figure 1A**). However, the LOA were wide for both groups at the commencement of the study for LA reservoir strain (competent LOA, -3.95%-5.43%; novice LOA, -7.32%-6.26%; **Figure 1B**).

A plateau was observed in both competent observers (competent ICC, 0.923; 95% CI, 0.891-0.993) and novice observers (novice ICC, 0.976; 95% CI, 0.910-0.994) after completing 50 cases (**Figure 1A**). Narrower LOA were also observed in both groups after completing 50 studies compared to expert observers (competent LOA, -1.86%-4.66%; novice LOA, -4.00%-2.88%; **Figure 1B**). No significant

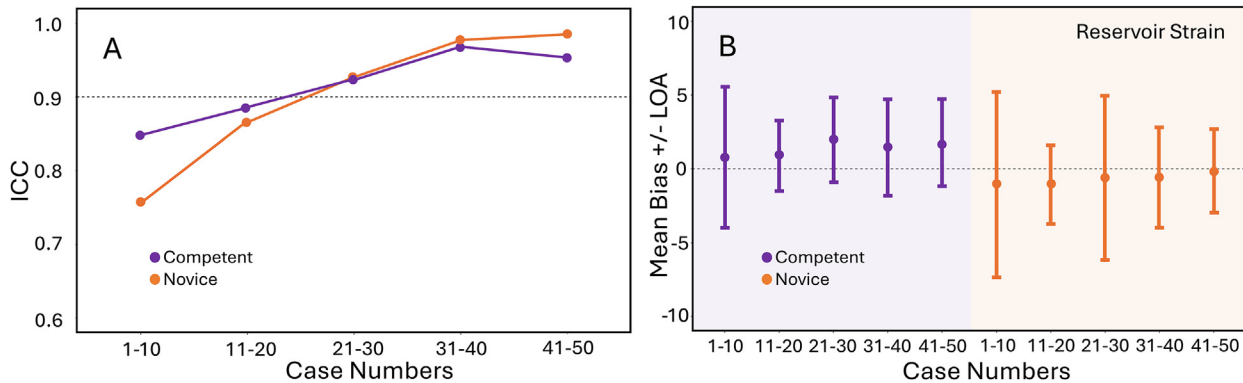


Figure 1 (A) Biplane LA reservoir strain learning curve. The ICC of LV strain competent and novice groups versus experts for each group of 10 studies. (B) Comparison of interobserver agreement of biplane LA reservoir strain between LV strain competent and novice groups versus experts (mean bias $\pm$ LOA).

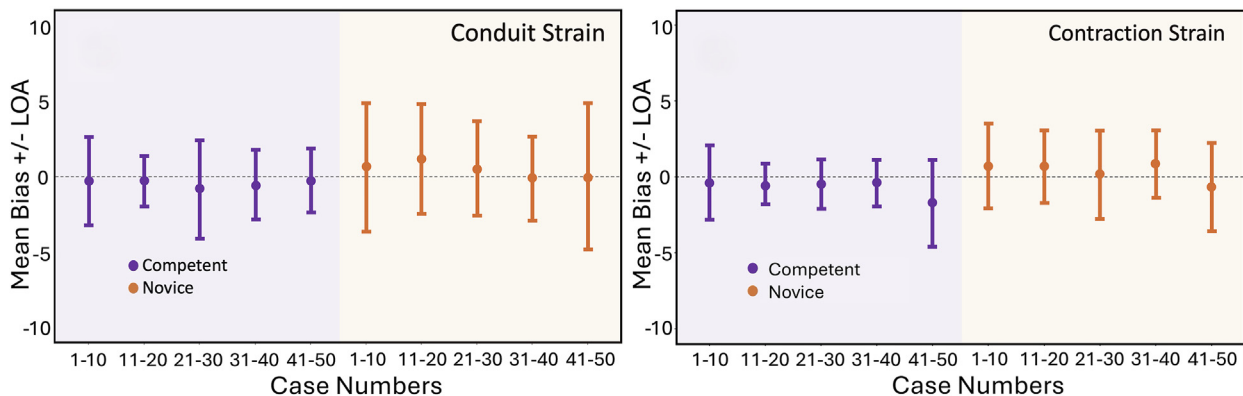


Figure 2 Comparison of interobserver agreement of biplane LA conduit and contraction strain between LV strain competent and novice groups versus experts (mean bias $\pm$ LOA).

differences in agreement and reproducibility were observed between both groups and experts once a plateau in learning curve was reached. For LA conduit and contraction strain, the competent group demonstrated superior levels of agreement with narrower LOA (Figure 2).

There was progressive improvement in time taken for LA strain analysis for both the novice group (156 ± 63 sec, $P < .001$) and the competent group (120 ± 20 sec, $P < .05$), with a significantly longer duration for the first 10 patients compared to experts (91 ± 17 sec). By the end of the study the average analysis time of the novice (93 ± 10 sec, $P = .19$) and competent group (99 ± 22 sec, $P = .16$) approached that of experts (87 ± 12 sec).

Limitations of this study included retrospective analysis with good image quality, which may predispose to selection bias. The number of observers in each group was small, but we ensured participants included both sonographers and physicians. There was more emphasis on LA reservoir strain analysis, but this has the greatest potential for clinical application. Our learning curve is applicable to a single vendor, but the same concept may be applicable to other vendors.

This study establishes that a significant learning curve exists for LA strain extending to individuals with prior strain experience who demonstrated a quicker learning curve. A minimum of 50 studies is recommended to achieve competency in diagnostic consistency, time efficiency, and reproducibility, similar to the GLS learning curve.⁵

CONFLICTS OF INTEREST

None.

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